RC Car Self-Driving Parts Plan

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This document organizes the current RC car hardware into four groups for a clean first self-driving build.

Keep

These parts should stay installed for version 1 of the self-driving platform.

- Raspberry Pi main computer
- Wide front USB webcam
- LiDAR
- Motor controller
- Drive motors
- Steering servo
- Main battery and power wiring
- Manual override controller or kill switch
- Core motor and steering control code

Reason: These are the minimum parts needed to sense the environment, steer, drive, and safely take over control.

Opt 1

These are the first optional parts worth keeping if you want extra safety and better control.

- Hall sensor
- Front ultrasonic sensor

Reason: - The hall sensor improves speed estimation. - The front ultrasonic sensor gives a short-range backup stop signal if vision or LiDAR misses something close.

Opt 2

These parts are useful later, but they are not needed for the first self-driving version.

- Right ultrasonic sensor
- Raspberry Pi Camera using Picamera2
- GPS module
- IR sensors

Reason: - The right ultrasonic sensor is mainly useful for curb or wall following. - The Pi camera is best saved for a rear or debug view after the main system works. - GPS is more useful for later outdoor waypoint navigation than for the first local self-driving stack. - IR sensors are not a priority compared with camera and LiDAR.

Remove

These parts can be removed now because they add wiring and software complexity without helping version 1 autonomy.

- Headlights
- Brake lights
- Left and right indicators
- Horn or buzzer
- MAX7219 LED display
- Voice assistant and speech code
- Photo capture and video recording extras
- Dashboard and UI extras not required for driving

Reason: These parts are convenience or cosmetic features, not core autonomy hardware.

Add Later After Self-Driving Works

These parts can be added back after the first self-driving build is stable.

- Pi camera as a rear camera or debug camera
- GPS for outdoor waypoint navigation
- Right ultrasonic sensor for wall-follow behavior
- Lights or simple status LEDs
- Logging and video recording
- GUI and telemetry display improvements

Recommended Build Order

1. Keep only the core drive, steering, compute, webcam, and LiDAR hardware.
2. Optionally keep the hall sensor and front ultrasonic sensor as safety helpers.
3. Remove lights, horn, display, and extra media or voice features.
4. Get the first self-driving loop working with the wide webcam and LiDAR.
5. Add optional sensors only after the base system is stable.